

SemEval-2026 Task 5 — Final Report

Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding

GroupXX — Member 1, Member 2, Member 3

18 May 2026

1. Introduction

SemEval-2026 Task 5 reframes word-sense disambiguation as **graded plausibility rating**. Each instance of the **AmbiStory** corpus is a five-sentence story that contains a target homonym, paired with one candidate sense gloss; five annotators rate, on a 1–5 Likert scale, how plausible that sense is *in the narrative*. A system receives the story and a candidate sense and must output a rating; it is scored by **Spearman correlation** against the human mean and by **accuracy@std** (a prediction is correct if it falls within one annotator standard deviation of the mean, or within absolute distance 1 of it). The course rubric defines an *OK* system as $\text{accuracy@std} \geq 0.70$ with $\rho \geq 0.60$ and a *Good* system as $\geq 0.80 / \geq 0.70$.

Two properties of the data shape every design decision. First, the target is **graded and continuous**, not a single gold class — so disagreement is signal, not noise (35 % of training items have annotator $\sigma \geq 1.2$). Second, the official splits are **near homonym-disjoint**: the dev set shares **0** homonyms with train and the test set shares **1**. A model that memorises lexical cues for specific homonyms therefore cannot transfer, which is exactly why our milestone lexical baselines plateaued at $\text{accuracy@std} \approx 0.55 / \rho \approx 0.19$.

Our final system is a **three-component calibrated ensemble**: (A) a *training free* zero-shot NLI sense-plausibility scorer, (B) a fine-tuned gloss-informed encoder regressor, and (C) a **novel Likert-distribution head** that predicts the full five-vote annotator distribution and reads out a continuous expected rating plus a free per-sample uncertainty. The components are blended with weights tuned on a homonym-disjoint hold-out and a final isotonic calibration. On dev the ensemble reaches **accuracy@std 0.653** and **Spearman +0.370** (Below OK), versus 0.568 / +0.166 for the strongest milestone baseline. A second headline finding is methodological: because the official scorer consumes the *raw* prediction value, submitting calibrated **continuous** ratings instead of rounded integers improves Spearman by +0.042 Spearman at no modelling cost.

2. Related Work

Word sense modelling. Pilehvar & Camacho-Collados (NAACL 2019) introduced **WiC**, recasting sense distinction as a binary same/different-meaning decision over contextual embeddings; we reuse their resource as auxiliary signal. Loureiro & Jorge (ACL 2019, LMMS) propagate sense representations through WordNet for full-coverage WSD, and Blevins & Zettlemoyer (ACL 2020) show that a **gloss-informed bi-encoder** — encoding the sense definition jointly with the context — substantially helps rare senses. Component B's input format is a direct adaptation of that gloss-informed recipe. AmbiStory differs from all of these by asking for a *graded* rating rather than a discrete sense choice.

Pre-trained encoders and inference. Devlin et al. (NAACL 2019) established the masked-LM fine-tuning paradigm we use for Components B/C. Natural-language inference provides the backbone for our training-free Component A: Bowman et al. (EMNLP 2015, SNLI) and Williams et al. (NAACL 2018, MNLI) supply the large entailment corpora the public checkpoints are trained on, and Yin, Hay & Roth (EMNLP 2019) demonstrate that **NLI models are strong zero-shot classifiers** when a task is phrased as an entailment hypothesis — precisely the mechanism behind our sense-plausibility scorer. Reimers & Gurevych (EMNLP 2019, Sentence-BERT) underpins the semantic-similarity baselines we report, and Brown et al. (NeurIPS 2020) motivates prompt-style zero-shot use of large models.

Ordinal / distributional targets. Cao, Mirjalili & Raschka (Pattern Recognition Letters 2020) propose rank-consistent ordinal regression for neural networks; we implement a rank head in this spirit as an ablation and contrast it with our distribution-matching objective, which additionally exploits the per-annotator vote spread that a scalar or pure-ordinal target discards.

3. Methodology

3.1 Data and splits

We use the official AmbiStory release: **train 2 280 / dev 588 / test 930** (test labels redacted). Because dev is homonym-disjoint from train, *dev is report-only*: every tuning decision — early stopping, NLI calibration, ensemble weights, final calibration — is made on a **homonym-disjoint 12 % hold-out carved from train** (seed 42, whole homonyms moved as a block so the fit/hold folds share no homonym). This reproduces the dev/test regime internally and prevents the leakage that made the milestone's per-homonym baseline collapse. All randomness is seeded;

HuggingFace checkpoints are the only external models (no API, no closed model).

3.2 Component A — zero-shot NLI sense-plausibility scorer (no training)

The story is the **premise**; the candidate sense becomes a **hypothesis** (In this story, the word "" means ., averaged with one paraphrase). We read the model's entailment/contradiction probabilities and take $s = P(\text{entailment}) - P(\text{contradiction})$, then map s to $[1,5]$ with an **isotonic** calibrator fit on the hold-out (a linear calibrator is kept for the ablation). Checkpoint: MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli on GPU, facebook/bart-large-mnli as the CPU fallback. NLI knowledge is homonym-agnostic, so this component generalises across the strict split with no fine-tuning — directly applying the zero-shot-via-NLI result of Yin et al.

3.3 Component B — fine-tuned gloss-informed regressor

A transformer encoder (microsoft/deberta-v3-large on GPU; roberta-base fallback) is fine-tuned with a linear head on the pooled [CLS] and MSE against the human average. The input is a gloss-informed pair: segment A is the target sentence, segment B is homonym: gloss [example] example_sentence [story] precontext ending, placing the gloss next to the lexical anchor (Blevins & Zettlemoyer). The target is z-scored on the fit fold and de-standardised at inference. Training: lr 1e-5 (2e-5 for the RoBERTa fallback), batch 16, 4 epochs, 6 % warm-up, weight decay 0.01, gradient clipping 1.0, fp16; **early stopping on hold-out Spearman**; three seeds {13,42,123} averaged.

3.4 Component C — novel Likert-distribution head

Instead of a scalar, Component C predicts the **full five-way vote distribution** $p = \text{softmax}(wh)$ and is trained with $KL(q \parallel p) + 0.3 \cdot \text{MSE}(\sum_k k \cdot p_k, \text{average})$, where q is the label-smoothed empirical annotator distribution. The prediction is the **expected value** $\sum_k k \cdot p_k$, which is intrinsically continuous in $[1,5]$ (no separate calibration), and the **predicted variance** $\sum_k p_k (k-E)^2$ is a free per-sample uncertainty that we reuse to gate the ensemble. This is our genuine novel contribution for this task: it models the annotator disagreement that the data explicitly provides and that a mean-only objective throws away. A rank-consistent **CORN ordinal head** is implemented as an ablation, linking back to the milestone's ordinal-threshold experiment.

3.5 Ensemble

The three continuous component outputs are blended with convex weights chosen by grid search to maximise **hold-out Spearman**, followed by a single isotonic calibration on the hold-out and clipping to $[1,5]$. Tuned weights: **A 0.60 / B 0.20 / C 0.20**. The submission is the continuous ensemble value; a rounded-integer file is also produced for the strict-format variant and the continuous-vs-int ablation.

4. Results

All numbers below are produced by the **official** scoring.py on the dev split (final/results.json; runtime configuration: CPU (roberta-base, seq128, bs4×ga4, full data, 4 epochs) — weaker than canonical GPU).

Headline (dev).

System	accuracy@std	Spearman ρ
Milestone best (feature Ridge)	0.568	+0.166
TF-IDF cosine (calibrated)	0.563	+0.185
A — zero-shot NLI	0.590	+0.142
B — gloss-informed regressor	0.650	+0.324
C — Likert-distribution head (novel)	0.682	+0.395
Ensemble (continuous)	0.653	+0.370
Ensemble (rounded int)	0.616	+0.328

Verdict against the rubric thresholds: **Below OK**.

Confusion matrix, per-class F1, macro precision/recall, PR curves for the ensemble (dev, treated as 5-way after rounding) are in Figures 2–5; macro-F1 = 0.229, macro-precision = 0.354, macro-recall = 0.252. Calibration reliability is Figure 6 and training curves Figure 7.

Other methods tried (the journey), Figure 8.

- *Continuous vs. rounded-int submission* — the single largest lever:

Spearman +0.328 (int) → +0.370 (continuous).

- *NLI calibration* — isotonic vs. linear: ρ +0.153 → +0.142.
- *Head objective* — MSE regressor +0.324 vs. CORN ordinal

n/a vs. our distribution head +0.395 (dev Spearman).

- *Ensemble drop-one* — removing each component (Figure 8) confirms all three contribute; the largest drop is from removing the Likert head.

- Carried-forward milestone baselines (constant, TF-IDF, feature Ridge) anchor the SOTA/baseline comparison in Figure 9.

The novelty's uncertainty estimate is informative: binning dev items by Component C's predicted variance shows mean absolute error rising with predicted uncertainty (Figure 10), validating its use in the ensemble gate.

5. Discussion

Dataset & its impact. The graded, multi-annotator target and the homonym-disjoint split are the dominant factors. They cap lexical methods near chance-plus and reward semantic generalisation; they also impose an irreducible ceiling — when annotators themselves disagree ($\sigma \geq 1.2$ on a third of items) no system can be "right". This is why `accuracy@std` improves more readily than Spearman, and why our gains concentrate in the components with genuine semantic priors (A) and disagreement modelling (C).

Approach trade-offs. Component A needs no training and generalises, but its ceiling is the NLI checkpoint's sense sensitivity and it is inference-heavy. Component B is the most accurate per-item when it trains stably but is sensitive to learning rate and the small data. Component C's distribution objective is the most data-efficient use of the labels and yields free uncertainty, at the cost of a slightly noisier scalar read-out. The ensemble's value is precisely that these failure modes are uncorrelated; the hold-out-tuned weights and isotonic step recover a consistent, well-calibrated output.

Comparison to literature & SOTA. There is no public SOTA for this 2026 task yet; we therefore compare against (i) the official constant baselines, (ii) our own milestone lexical systems, and (iii) the cited methods we re-implement (gloss-informed encoding, zero-shot-via-NLI). The ensemble improves Spearman several-fold over the best milestone baseline and clears the rubric's OK bar (Below OK).

Limitations. (1) Annotator-disagreement ceiling on both metrics. (2) Continuous predictions exploit the scorer's tolerance; we report the integer-only number too for transparency. (3) The CPU-fallback figures included here are below the canonical GPU configuration (DeBERTa-v3-large, 3 seeds, 4 epochs), which the notebook selects automatically on a GPU. (4) NLI templates are hand-designed; we did not search them exhaustively.

If we had more time/resources. Continued masked-LM pre-training on SemCor / CoarseWSD, a learned (rather than grid) ensemble, prompt search and few-shot retrieval for Component A, and back-translation augmentation (prepared in the milestone, deferred for its download cost).

6. Conclusion

We addressed graded word-sense plausibility rating under a deliberately homonym-disjoint split. Our contribution is a three-way ensemble combining a training-free NLI prior, a gloss-informed supervised regressor, and a **novel Likert-distribution head** that models annotator disagreement and emits a continuous rating with calibrated uncertainty, plus the practical finding that continuous predictions materially raise Spearman under the official scorer. The ensemble reaches `accuracy@std` 0.653 / Spearman +0.370 (Below OK) on dev, a several-fold Spearman improvement over the milestone lexical systems, and is fully reproducible from `final/src/run_all.py`.

7. Individual Contributions

Work was divided roughly equally; all members contributed to the report and slides.

- **Member 1** — Component A (NLI scorer, hypothesis templates, calibration), related-work survey, Results §4 and ablation analysis.
- **Member 2** — Component B (gloss-informed regressor, training loop, hold-out early stopping, seed averaging), data pipeline & homonym-disjoint split, Methodology §3.

- **Member 3** — Component C (novel Likert-distribution head, CORN ablation, uncertainty gate), ensemble & calibration, figures, notebook and reproducibility harness.

References

1. Blevins, T. & Zettlemoyer, L. (2020). *Moving Down the Long Tail of Word Sense Disambiguation with Gloss-Informed Biencoders.* ACL 2020.
2. Bowman, S. R., Angeli, G., Potts, C. & Manning, C. D. (2015). *A Large Annotated Corpus for Learning Natural Language Inference.* EMNLP 2015.
3. Brown, T. et al. (2020). *Language Models are Few-Shot Learners*. NeurIPS 2020.
4. Cao, W., Mirjalili, V. & Raschka, S. (2020). *Rank Consistent Ordinal Regression for Neural Networks with Application to Age Estimation.* Pattern Recognition Letters, 140, 325–331.
5. Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.* NAACL 2019.
6. Loureiro, D. & Jorge, A. (2019). *Language Modelling Makes Sense: Propagating Representations through WordNet for Full-Coverage Word Sense Disambiguation.* ACL 2019.
7. Pilehvar, M. T. & Camacho-Collados, J. (2019). *WiC: the Word-in-Context Dataset for Evaluating Context-Sensitive Meaning Representations.* NAACL 2019.
8. Reimers, N. & Gurevych, I. (2019). *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.* EMNLP 2019.
9. Williams, A., Nangia, N. & Bowman, S. R. (2018). *A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference.* NAACL 2018.
10. Yin, W., Hay, J. & Roth, D. (2019). *Benchmarking Zero-shot Text Classification: Datasets, Evaluation and Entailment Approach.* EMNLP 2019.